

Rate of Whole-Breast Hypofractionated Radiation Therapy Before Versus After Virtual Tumor Board Implementation

Adam C. Powell, PhD¹; Mitchell Ceylan, BBA¹; James W. Long, BSBA²; Uday U. Deshmukh, MD, MPH¹; Bryan Loy, MD, MBA²; and Amin J. Mirhadi, MD¹

QUESTION ASKED: Did the rate of whole-breast hypofractionated (HF) radiation therapy (RT) increase after a virtual tumor board (VTB) was implemented?

SUMMARY ANSWER: A significantly higher percentage of orders and claims were for HF RT after the VTB was implemented, relative to what had been observed before it was implemented.

WHAT WE DID: The VTB is a multidisciplinary team of 15 board-certified physicians with expertise in RT, chemotherapy, and surgical oncology—six of whom are specialists in breast cancer. The physicians on the VTB practice medicine at a variety of facilities across the country and are part-time contractors of the specialty benefits management company operating the VTB. During quarterly meetings, the members of the VTB develop a consensus on the types of treatment plans that could be provided to physicians. When a physician issues an order for a patient indicative of a cancer diagnosis that may benefit from a treatment-planning discussion with the VTB, a physician affiliated with the VTB will reach out to the physician to offer assistance within roughly a day of the order. As a result of the National Comprehensive Cancer Network guidelines on the treatment of breast cancer after breast-conserving surgery, which recommend the use of HF RT as a preferred treatment approach, the VTB has a policy of encouraging the inclusion of HF RT in treatment plans, when appropriate. An observational, cohort analysis using prior authorization and claims data

were conducted to show how the relative use of HF and conventional-fractionated RT changed after the implementation of the VTB.

WHAT WE FOUND: In the year after implementation of the VTB, we observed a significantly higher percentage of orders for HF RT (60.3% [n = 1,254 of 2,079] v 53.2% [n = 1,010 of 1,899]; $P < .001$) and claims for HF RT (71.5% [n = 1,143 of 1,598] v 59.0% [n = 941 of 1,595]; $P < .001$). Relative to the year prior to implementation of the VTB, the adjusted odds of an order for HF RT was 1.35 (95% CI, 1.19 to 1.54), and the adjusted odds of a claim for HF RT was 1.76 (95% CI, 1.52 to 2.04).

BIAS, CONFOUNDING FACTORS: Despite the strength of the findings, they constitute an observed association, so it is possible that they were driven by a factor other than the initiation of the VTB. There may have been contemporaneous changes in the use of HF RT caused by the growing awareness of its benefits during the time of the study. The findings of this study may not be generalizable to other contexts. The study examined one type of cancer, one organization's implementation of a VTB, and patients with health plans from one national organization.

REAL-LIFE IMPLICATIONS: When addressing the needs of patients, particularly those who may have limited access to specialist consultations, health plans and physicians may wish to consider the potential impact of a VTB on the quality of care.

Author affiliations and disclosures are available with the complete article at ascopubs.org/journal/op.

Accepted on August 12, 2020 and published at ascopubs.org/journal/op on October 8, 2020: DOI <https://doi.org/10.1200/JOP.20.00480>

CORRESPONDING AUTHOR

Adam C. Powell, PhD, 16945 Northchase Dr, Suite 1300, Houston, TX 77060; Twitter: @healthhelp; e-mail: powellad@healthhelp.com.

Rate of Whole-Breast Hypofractionated Radiation Therapy Before Versus After Virtual Tumor Board Implementation

Adam C. Powell, PhD¹; Mitchell Ceylan, BBA¹; James W. Long, BSBA²; Uday U. Deshmukh, MD, MPH¹; Bryan Loy, MD, MBA²; and Amin J. Mirhadi, MD¹

abstract

PURPOSE The virtual tumor board (VTB) is a multidisciplinary group of specialist physicians who remotely educate the treating physician on the development of an evidence-based cancer treatment plan that will enhance patient outcomes according to the available literature. The use of hypofractionated (HF) radiation therapy (RT) is a preferred approach according to National Comprehensive Cancer Network guidelines and is encouraged by the VTB, when appropriate.

MATERIALS AND METHODS An observational, cohort study using prior authorization and claims data were conducted to show how the relative use of HF and conventional fractionated (CF) RT changed after the implementation of the VTB. Orders and claims for qualifying patients from 1 year before launch (August 2016) to 1 year after launch (August 2018) of the VTB were extracted. Claims were examined to observe which patients received CF (28-35 fractions) versus HF (15-21 fractions) RT. χ^2 tests were used to assess the association between time period and the ordering and use of HF RT. Logistic regressions were used to test the association, after adjusting for the patient's age, urbanicity, local average income, and the RT modality used.

RESULTS After implementation, we observed a significantly higher percentage of orders for HF RT (60.3% [n = 1,254 of 2,079] v 53.2% [n = 1,010 of 1,899]; $P < .001$) and claims for HF RT (71.5% [n = 1,143 of 1,598] v 59.0% [n = 941 of 1,595]; $P < .001$). Relative to before implementation, the adjusted odds of an order for HF RT was 1.35 (CI, 1.19 to 1.54), and the adjusted odds of a claim for HF RT was 1.76 (CI, 1.52 to 2.04).

CONCLUSION After the VTB was implemented, there was a significant increase in HF RT orders and claims.

JCO Oncol Pract 16. © 2020 by American Society of Clinical Oncology

INTRODUCTION

The treatment of breast cancer often requires patients to have repeated encounters with the health care system over a period of several months. As a result, patients may seek treatment locally to simplify their logistics. The resources available locally to patients are highly variable, with some patients having greater access to specialists than others. Access to specialists is particularly problematic in rural areas, and virtual tumor boards (VTBs) have been pursued as one potential solution.¹

To improve the quality of breast cancer care and ensure uniform access to expertise, one national health care organization has implemented a VTB program for its health insurance plan members. The VTB works telephonically with a patient's treating physician to develop a nonbinding, evidence-based

treatment plan. Telephonic discussion with the VTB provides the treating physician the opportunity to consult specialists who may not be otherwise available within the organization delivering the care. Though treating physicians are not required to follow the treatment plan that they develop with the VTB and may need to deviate from it if the patient's trajectory changes, orders congruent with the treatment plan are automatically approved by the specialty benefits management company's nondenial prior authorization program.

As a result of the National Comprehensive Cancer Network (NCCN) guidelines on the treatment of breast cancer after breast-conserving surgery, which recommend the use of hypofractionated (HF) radiation therapy (RT) as a preferred treatment approach, the VTB has a policy of encouraging the inclusion of HF RT in treatment plans, when appropriate.² There is

Author affiliations and support information (if applicable) appear at the end of this article.

Accepted on August 12, 2020 and published at ascopubs.org/journal/op on October 8, 2020; DOI <https://doi.org/10.1200/OP.20.00480>

substantial clinical evidence supporting the use of HF RT in the treatment of breast cancer, most notably the START-A and START-B clinical trials.^{3,4}

Although VTBs have previously been shown to be feasible and accepted by physicians, there has not been prior research on their impact on breast cancer care.⁵⁻⁸ To fill this gap in the literature, this study examined whether orders and claims for RT for breast cancer shifted from conventional fractionation (CF) toward HF RT after the implementation of a VTB. Furthermore, this study demonstrates the performance of a VTB implemented nationally, with a multi-institutional set of practicing physicians.

MATERIALS AND METHODS

Description of the VTB Intervention

The VTB is a multidisciplinary team of 15 board-certified physicians with expertise in RT, chemotherapy, and surgical oncology—six of whom are specialists in breast cancer. The physicians on the VTB practice medicine at a variety of facilities across the country and are part-time contractors of the specialty benefits management company operating the VTB. During quarterly meetings, the members of the VTB develop a consensus on the types of treatment plans that could be provided to physicians.

When a physician issues an order for a patient indicative of a cancer diagnosis that may benefit from a treatment-planning discussion with the VTB, a physician affiliated with the VTB will reach out to the physician to offer assistance within roughly a day of the order. Discussing the order with the VTB is voluntary, and physicians that elect to opt out still receive authorization for their orders. Nonetheless, the majority of physicians have historically agreed to participate. All of the information needed for treatment planning is obtained through the discussion between the treating physician and the physician reaching out on behalf of the VTB; no information is extracted from an electronic health record. During the discussion, a treatment plan is developed by considering professional guidelines, the scientific literature, the discussion among the tumor board, and the unique circumstances of the patient's presentation. Often, the treatment plans for three to five of the physician's patients are discussed during the session. As needed, outreach is made to the other specialists on the VTB for input.

Data Source and Sample Population

All prior authorization orders for RT for the treatment of breast cancer with dates of service between August 1, 2016 and July 31, 2018, pertaining to patients outside the market where the VTB was piloted, were extracted from the database of a company performing nondenial prior authorization for radiation oncology. The nondenial prior authorization program provided treating physicians with pretreatment consultations with nonbinding recommendations in situations in which orders could not be immediately approved. The RT

orders all pertained to patients with health insurance plans from one national health care organization. The date of the RT order served as the index date. Orders were excluded from the study if they pertained to patients age ≥ 90 years, with a history of mastectomy, with positive lymph nodes, with health plan disenrollment within 10 weeks after the order, or with a listed clinical indication ineligible for the VTB intervention. Although all orders that met the inclusion and exclusion criteria could have been reviewed by the VTB in the postimplementation period, only a subset of the orders in the postintervention period were actually reviewed due to capacity constraints. Unreviewed orders in the postintervention period were included in the analysis.

Using the population of patients created with the inclusion and exclusion criteria, two separate subpopulations were created. To derive the population for the order analysis, an additional exclusion was made for patients whose orders did not meet the definition of HF (15-21 fractions) or CF (28-35 fractions) RT. To derive the population used for the claims analysis, additional exclusions were made for patients who had RT claims > 10 weeks after their order and for patients whose claims did not meet the definition of HF or CF RT. Note that the number of fractions designated in an order did not always match the number of fractions found in the claims data for the corresponding patient: patients may have discontinued treatment, or physicians may have opted to modify treatment (increasing or decreasing the number of fractions) as treatment continued.

This study was reviewed by Advarra (Columbia, MD) institutional review board (IRB) and received an exemption from IRB oversight, in accordance with the Department of Health and Human Services regulations found at 45 CFR 46.104(d)(4).

Measurement

The dependent variable in the analyses were whether the patient had an order for CF or HF RT and whether the patient had claims indicating that CF or HF RT had been delivered. The independent variable in the analysis was whether the patient's order for RT was placed in the year before the VTB was initiated (August 1, 2016 to July 31, 2017) or in the year after the VTB was initiated (August 1, 2017 to July 31, 2018).

The covariables included as potential confounders in the analysis were variables for whether patients lived in areas with average incomes $< \$40,000$, sex, urbanicity, health plan line of business (commercial v Medicare), treatment modality, and age. Average income within a patient's ZIP code was determined using the American Community Survey's 2013-2017 5-year estimates of median income, which reported income in 2017 inflation-adjusted dollars.⁹ Rural areas were identified using a ZIP code mapping table developed by the Centers for Medicare & Medicaid Services.¹⁰

Analysis

Descriptive statistics were calculated for the sample of patients in the order analysis and for the sample of patients in the claims analysis. Using χ^2 tests, the patients receiving orders before the initiation of the VTB were compared with those receiving orders after the initiation of the VTB on each of the covariables. This process was repeated for the claims analysis.

Logistic regressions were run to determine the adjusted association between the timing of the order and the ordering of HF RT, and the timing of the order and the presence of claims for HF RT. Results from the logistic regressions were reported as odds ratios.

RESULTS

As shown in [Figure 1](#), of the 5,299 patients meeting the original inclusion criteria, 3,978 were ultimately included in the order analysis, and 3,193 were ultimately included in the claims analysis. As is shown in the descriptive statistics analysis ([Table 1](#)), the orders pertaining to the pre- versus post-VTB populations did not significantly differ on any of the covariables except health plan; compared with the pre-VTB population (88.6% of the patients enrolled in Medicare health plans), the post-VTB population had 90.8% of patients participating in Medicare health plans ($P < .03$). The descriptive statistics for the claims analysis likewise did not find any significant differences between the pre- and post-VTB populations other than health plan type. Within the claims analysis, 89.1% of the pre-VTB population and 92.6% of the post-VTB population had Medicare health plans ($P < .001$).

In each of the subpopulations examined, the average age of the patient was approximately 69 years. Before the VTB, 3.4% of orders were from women < 50 years; after the VTB, women < 50 years accounted for 2.5% of orders, which was an insignificant difference ($P = .104$). Meanwhile, before the VTB, 3.1% of claims were from women < 50 years, whereas women < 50 years accounted for 1.9% of claims after the VTB, which was a significant difference ($P = .031$).

After the VTB was implemented, a significantly higher percentage of orders (60.3% v 53.2%; $P < .001$) and a significantly higher percentage of claims (71.5% v 59.0%; $P < .001$) were for HF RT. In the order analysis, orders made after the VTB had 1.35 adjusted odds of being for HF RT (95% CI, 1.19 to 1.54). Likewise, in the claims analysis, claims pertaining to orders made after the VTB had 1.76 adjusted odds (95% CI, 1.52 to 2.04) of being for HF RT. Although the odds ratios slightly differed between the order analysis and the claims analysis ([Table 2](#)), the same factors were significantly associated with CF versus HF RT, and the directions of the associations were the same. Patients were significantly less likely to have orders and claims for HF RT if they had their order placed before

the initiation of the VTB, were of a younger age, resided in a ZIP code with an average income of $< \$40,000$ per year, had Medicare Advantage, and/or had intensity-modulated RT as the designated modality.

DISCUSSION

After the implementation of the VTB, patients receiving whole-breast RT were more likely to receive both orders and claims for HF RT. The pre-VTB and post-VTB populations of patients were similar, so this difference suggests that the VTB may have increased the likelihood that patients received orders for HF RT and ultimately had claims for HF RT. The patients received their cancer care at facilities across the country, helping demonstrate that a VTB can be implemented nationally. Thus, these findings provide an additional point of evidence supporting the acceptability and feasibility of VTBs in diverse practice settings, bolstering what has been found previously.⁵⁻⁸

Evidence supporting the use of HF RT in the treatment of whole-breast RT in women age ≥ 50 years began to accumulate in 2010, and more evidence supporting its use in women < 50 years became available in 2013.^{3,4} Guidelines from the American Society for Radiation Oncology supporting the use of HF RT in the treatment of older women were released in 2011 and were updated to support the use of HF RT in the treatment of all ages in March 2018, during this study period.^{11,12} Although the evolution of the guidelines may have had the potential to affect the findings of this analysis, women < 50 years were in the minority of those included in the analysis. Other researchers have found that HF whole-breast RT has remained underused, even in women ≥ 50 years.¹³

It is possible that the increased use of HF RT observed in the post-VTB period was reflective of a national trend toward HF RT adoption rather than the impact of the initiation of the VTB, so a follow-on analysis was conducted to better understand contemporaneous trends. Before it was implemented nationally, the VTB was available to Florida-based patients with Medicare health plans from the organization as a pilot for the program. These patients were excluded from the primary analysis, because they had access to the VTB during both periods of the study. This pilot population did not experience a significant change in its ordering of HF RT ($P = .79$) or claims ($P = .98$) for HF RT when orders and claims from August 1, 2016, to July 31, 2017, were compared with those from August 1, 2017, to July 31, 2018. These findings suggest that time trends alone, absent a programmatic change, may not account for the significant changes in HF RT ordering and claims that were observed in the national population.

The VTB has the potential to improve the quality of care through fostering the use of HF RT, the clinical benefits of which have been established by multiple clinical trials.^{3,4} Consequently, HF RT has been promoted in clinical

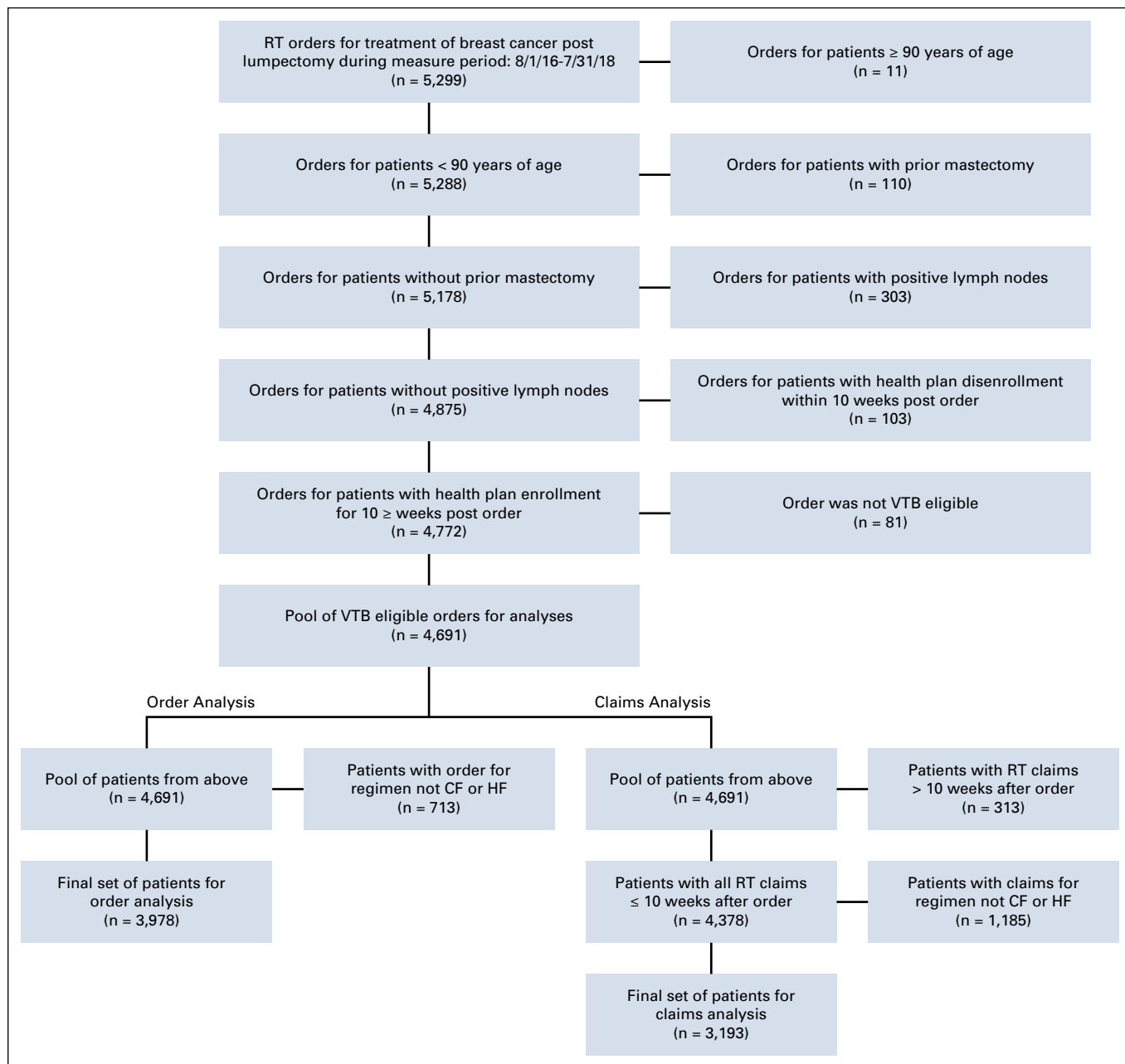


FIG 1. Participant flow diagram. CF, conventional fractionation; HF, hypofractionation; RT, radiation therapy; VTB, virtual tumor board.

practice guidelines.¹ Furthermore, prior research has shown there is an association between the selection of an HF RT regimen and treatment completion, in the context of patients receiving RT after mastectomy.¹⁴ Although the patient population slightly differed from the post-lumpectomy population examined by this study, this finding suggests that the VTB's efforts to promote HF RT may have positive implications for treatment completion.

It has been noted by prior researchers that uptake of HF RT in the United States has been slow relative to that seen in other nations and that the incentive structure of the American health care system generally has not fostered the

use of HF RT.^{15,16} Nonetheless, the Radiation Oncology Advanced Payment Model proposed by Congress has the potential to make physicians economically indifferent to the length of RT provided by compensating providers on the basis of episode-based payment, rather than fee-for-service payment.¹⁷ However, this shift may not be enough to drive increased hypofractionation. The health care organization that is the subject of this study began paying a small number of treatment facilities on a case rate basis in 2013. When fee-for-service payment was in effect in 2011, patients with breast cancer received an average of 29.7 fractions of RT. In contrast, when case rate payment was in

TABLE 1. Descriptive Statistics

Variable	Order Analysis				Claims Analysis			
	All (N = 3,978)	Pre VTB (n = 1,899)	Post VTB (n = 2,079)	P	All (N = 3,193)	Pre VTB (n = 1,595)	Post VTB (n = 1,598)	P
Modality: IMRT v 2D3D	41 (1.0)	25 (1.3)	16 (0.8)	.088	35 (1.1)	23 (1.4)	12 (0.8)	.061
Health plan: Medicare v commercial	3,570 (89.7)	1,683 (88.6)	1,887 (90.8)	.030	2,900 (90.8)	1,421 (89.1)	1,479 (92.6)	< .001
Urbanicity: Rural v urban or omitted	761 (19.1)	367 (19.3)	394 (19.0)	.795	629 (19.7)	312 (19.6)	317 (19.8)	.879
Income < \$40,000 v ≥ \$40,000 or omitted	877 (22.0)	410 (21.6)	467 (22.5)	.532	699 (21.9)	344 (21.6)	355 (22.2)	.689
Mean (SD) age, years	68.9 (8.1)	68.8 (8.3)	68.9 (8.0)	.648	69.1 (7.8)	68.9 (8.1)	69.3 (7.4)	.171

NOTE. Data are presented as No. (%) unless otherwise noted (ie, age variable).

Abbreviations: 2D3D, two- or three-dimensional conformal radiation therapy; IMRT, intensity-modulated radiation therapy; SD, standard deviation; VTB, virtual tumor board.

effect in 2013, patients with breast cancer received an average of 30.4 fractions of RT, which is a statistically insignificant increase.¹⁸ This finding, that the number of fractions increased rather than declined in response to case rates, suggests that the Radiation Oncology Advanced Payment Model on its own may not be enough to drive the use of HF RT. Interventions like VTBs may continue to play an important role in driving the utilization of HF RT, even when payments are made on a case rate basis.

Other researchers have also examined the impact of interventions for increasing the uptake of HF RT. The University of Kansas Medical Center, along with its community-based satellites, implemented a systemwide peer review process to foster the use of HF RT in patients treated for early-stage breast cancer. The rate of HF RT increased from 66% in January 2016 (before implementation) to 81% in June 2017 across the entire cancer care network ($P < .001$) and from 51% to 69% when the community-based satellites were considered alone ($P = .007$). This finding shows that similar rates of HF RT were achieved in the community surrounding one academic medical center by fostering dialogue like that promoted by the VTB.¹⁹

Despite the strength of the findings, they constitute an observed association, so it is possible that the results were driven by a factor other than the initiation of the VTB. There may have been contemporaneous changes in the use of HF RT because of the growing awareness of its benefits during the time of the study.¹³ Nonetheless, the short timeline of the study helps reduce the potential for changes in practice patterns to be responsible for the findings. Likewise, use of HF RT may have increased as a result of the sentinel effect, which has been observed elsewhere in the context of specialty benefits management, rather than any particular actions taken by the VTB.²⁰

Not all VTB-eligible orders in the post-VTB period underwent VTB review because of capacity constraints. These unreviewed orders were included in the analysis. This limitation biases the study away from having significant findings, because the VTB may have had a larger impact on care if it had reviewed all eligible orders. This limitation may account for the relatively modest increase in HF RT utilization compared to that observed elsewhere.¹⁹ Nevertheless, the education that treating physicians gained from interacting with the VTB may have affected their practice

TABLE 2. Adjusted Odds Ratios for Factors Potentially Influencing Regimen

Variable	Order Analysis		Claims Analysis	
	OR	95% CI	OR	95% CI
Timing of order: Post v pre VTB	1.35	1.19 to 1.54	1.76	1.52 to 2.04
Modality: IMRT v 2D3D	0.38	0.19 to 0.72	0.37	0.18 to 0.73
Health plan: Medicare v commercial	0.71	0.54 to 0.92	0.60	0.44 to 0.83
Urbanicity: Rural v urban or omitted	1.03	0.87 to 1.22	0.95	0.79 to 1.15
Income < \$40,000 v ≥ \$40,000 or omitted	0.79	0.68 to 0.93	0.78	0.65 to 0.93
Age, years	1.05	1.04 to 1.07	1.03	1.02 to 1.04

Abbreviations: 2D3D, two- or three-dimensional conformal radiation therapy; IMRT, intensity-modulated radiation therapy; OR, odds ratio; VTB, virtual tumor board.

patterns for orders that were not reviewed by the VTB. Furthermore, the VTB used its discretion and did not recommend HF RT in all cases. Patient cases were included regardless of whether HF RT was recommended by the VTB or not, because it would not be possible to exclude a comparable set of patient cases in the pre-VTB period.

The findings of this study may not be generalizable to other contexts. The study examined one type of cancer, one organization's implementation of a VTB, and patients with health plans from one national organization. The organization providing the health plans operates predominantly in the South and primarily provides Medicare Advantage health plans. The sample did not contain any patients with traditional Medicare (government-administered) health plans or with a Medicaid plan lacking dual eligibility for Medicare. Future researchers may wish to examine the

performance of VTBs in the context of multiple forms of cancer, multiple implementations of VTBs, and multiple health plan providers.

In conclusion, after the implementation of the VTB, patients were significantly more likely to have orders and claims for HF, as opposed to CF, RT. Although greater use of HF RT was observed in the post-VTB period, it is possible that local trends or other factors not considered in the analysis accounted for the difference. When addressing the needs of patients, particularly those who may have limited access to specialist consultations, health plans and physicians may wish to consider the potential impact of a VTB on the quality of care. Additional research is needed to determine the relationship between the implementation of a VTB and clinical outcomes.

AFFILIATIONS

¹HealthHelp, Houston, TX

²Humana, Louisville, KY

CORRESPONDING AUTHOR

Adam C. Powell, PhD, 16945 Northchase Dr, Suite 1300, Houston, TX 77060; Twitter: @healthhelp; e-mail: powellad@healthhelp.com.

SUPPORT

Supported by HealthHelp and Humana in the form of general compensation provided to the authors for participation in research activities as a part of their job responsibilities, rather than this particular study.

AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

Disclosures provided by the authors are available with this article at DOI <https://doi.org/10.1200/OP.20.00480>.

AUTHOR CONTRIBUTIONS

Conception and design: All authors

Collection and assembly of data: Mitchell Ceylan

Data analysis and interpretation: Adam C. Powell, Mitchell Ceylan, Uday U. Deshmukh

Provision of study material or patients: James W. Long, Uday U. Deshmukh, Amin J. Mirhadi

Administrative support: Adam C. Powell, Mitchell Ceylan, Uday U. Deshmukh

Manuscript writing: All authors

Final approval of manuscript: All authors

Accountable for all aspects of the work: All authors

REFERENCES

- Charlton M, Schlichting J, Chioreso C, et al: Challenges of rural cancer care in the United States. *Oncology (Williston Park)* 29:633-640, 2015
- Salerno KE: NCCN guidelines update: Evolving radiation therapy recommendations for breast cancer. *J Natl Compr Canc Netw* 15:682-684, 2017
- Haviland JS, Owen JR, Dewar JA, et al: The UK Standardisation of Breast Radiotherapy (START) trials of radiotherapy hypofractionation for treatment of early breast cancer: 10-year follow-up results of two randomised controlled trials. *Lancet Oncol* 14:1086-1094, 2013
- Whelan TJ, Pignol JP, Levine MN, et al: Long-term results of hypofractionated radiation therapy for breast cancer. *N Engl J Med* 362:513-520, 2010
- Marshall CL, Petersen NJ, Naik AD, et al: Implementation of a regional virtual tumor board: A prospective study evaluating feasibility and provider acceptance. *Telemed J E Health* 20:705-711, 2014
- Madhavan S, Blais EM, Bender RJ, et al: A cloud-based virtual tumor board to facilitate treatment recommendations for patients with advanced cancers. *J Clin Oncol* 36 (suppl; abstr 6508), 2018
- Stevenson MM, Irwin T, Lowry T, et al: Development of a virtual multidisciplinary lung cancer tumor board in a community setting. *J Oncol Pract* 9:e77-e80, 2013
- Salami AC, Barden GM, Castillo DL, et al: Establishment of a regional virtual tumor board program to improve the process of care for patients with hepatocellular carcinoma. *J Oncol Pract* 11:e66-e74, 2015
- Census.gov: S1903: Median income in the past 12 months (in 2017 inflation-adjusted dollars)—2013-2017 American community survey 5-year estimates. <https://data.census.gov/cedsci/table?q=S1903&tid=ACST1Y2017.S1903>
- Centers for Medicare & Medicaid Services: DME-rural-zip-and-formats. <https://www.cms.gov/Medicare/Medicare-Fee-for-Service-Payment/DMEPOSFeeSched/DMEPOS-Fee-Schedule-Items/DME-Rural-Zip-and-Formats.html?DLPage=1&DLEntries=10&DLSort=2&DLSortDir=descending>
- Smith BD, Bentzen SM, Correa CR, et al: Fractionation for whole-breast irradiation: An American Society for Radiation Oncology (ASTRO) evidence-based guideline. *Int J Radiat Oncol Biol Phys* 81:59-68, 2011
- Smith BD, Bellon JR, Blitzblau R, et al: Radiation therapy for the whole breast: Executive summary of an American Society for Radiation Oncology (ASTRO) evidence-based guideline. *Pract Radiat Oncol* 8:145-152, 2018
- Rodríguez-López JL, Ling DC, Heron DE, et al: Lag time between evidence and guidelines: Can clinical pathways bridge the gap? *J Oncol Pract* 15:e195-e201, 2019

14. Powell AC, Rogstad TL, Jacob NM, et al: The association between use of hypofractionation and treatment completion among recipients of radiation therapy post-mastectomy. *Pract Radiat Oncol* 10:e244-e249, 2020
15. Culbert M, De Los Santos JF, Saito AI, et al: Uptake of whole-breast hypofractionation: A rapid review of the literature. *Int J Radiat Oncol Biol Phys* 102:e575, 2018.
16. Bekelman JE, Sylwestrzak G, Barron J, et al: Uptake and costs of hypofractionated vs conventional whole-breast irradiation after breast conserving surgery in the United States, 2008-2013. *JAMA* 312:2542-2550, 2014
17. Gupta A, Ohri N, Haffty BG: Hypofractionated whole breast irradiation is cost-effective—But is that enough to change practice? *Transl Cancer Res* 7: S469-S472, 2018
18. Loy BA, Shkedy CI, Powell AC, et al: Do case rates affect physicians' clinical practice in radiation oncology? An observational study. *PLoS One* 11:e0149449, 2016
19. Ganju RG, TenNapel M, Chen AM, et al: Impact of peer review on use of hypofractionated regimens for early-stage breast cancer for patients at a tertiary care academic medical center and its community-based affiliates. *J Oncol Pract* 15:e153-e161, 2019
20. Friedman DP, Smith NS, Bree RL, et al: Experience of an academic neuroradiology division participating in a utilization management program. *J Am Coll Radiol* 6:119-124, 2009



AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST**Rate of Whole-Breast Hypofractionated Radiation Therapy Before Versus After Virtual Tumor Board Implementation**

The following represents disclosure information provided by authors of this manuscript. All relationships are considered compensated unless otherwise noted. Relationships are self-held unless noted. I = Immediate Family Member, Inst = My Institution. Relationships may not relate to the subject matter of this manuscript. For more information about ASCO's conflict of interest policy, please refer to www.asco.org/rwc or ascopubs.org/op/authors/author-center.

Open Payments is a public database containing information reported by companies about payments made to US-licensed physicians ([Open Payments](#)).

Adam C. Powell

Employment: HealthHelp/WNS, Payer+Provider Syndicate

Leadership: Payer+Provider Syndicate

Stock and Other Ownership Interests: Berkshire Hathaway, Community Health Systems, CVS Health, HCA Healthcare, Payer+Provider Syndicate, Quorum Health, and Tenet Healthcare

Mitchell Ceylan

Employment: HealthHelp/WNS

Uday U. Deshmukh

Employment: HealthHelp/WNS

Stock and Other Ownership Interests: WNS Holdings, Cigna, Merck, Pfizer, Procter & Gamble, Johnson & Johnson

Honoraria: PCORI, STOP - JIA

Consulting or Advisory Role: WNS

Bryan Loy

Employment: Humana

Stock and Other Ownership Interests: Humana

Research Funding: Humana (Inst)

Amin J. Mirhadi

Employment: HealthHelp/WNS

Consulting or Advisory Role: Varian

No other potential conflicts of interest were reported.